

# Mend-Amar Badral

 MendeBadra |  Mend-Amar Badral |  mendebadra.github.io |  
 mendamar.badr@gmail.com |  
 +36 70 7385 744

## EDUCATION

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|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------|
| <b>Budapest University of Technology and Economics (BME)</b><br>M.S. in Computer Science Engineering<br><i>Focus: Data Science &amp; Artificial Intelligence</i>  | Sep, 2025 – June, 2027<br>Budapest, Hungary     |
| <b>National University of Mongolia (NUM)</b><br>B.S. in Applied Mathematics<br>Thesis: <a href="#">Evaluating land degradation using image processing methods</a> | Sep, 2021 – June, 2025<br>Ulaanbaatar, Mongolia |

## EXPERIENCE

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| <b>Center of Mathematics Application Research Laboratory</b><br><i>Research Data Intern</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Oct, 2023 – June, 2025 |
| <ul style="list-style-type: none"><li>– <b>Automated Data Processing:</b> Engineered a Python-based image processing pipeline to ingest and process large datasets of drone imagery. Automated the classification of plant types (grass, weeds, soil) using efficient segmentation algorithms (NumPy, OpenCV).</li><li>– <b>Software Engineering &amp; OOP:</b> Refactored legacy research code into a modular, object-oriented (OOP) architecture. Improved code maintainability and reduced technical debt, aligning with standard software engineering practices.</li><li>– <b>Data Validation:</b> Implemented statistical methods (Moran's I) to validate spatial data quality, achieving a detection accuracy confidence of 0.58–0.70 for weed growth patterns.</li><li>– <b>Technical Documentation:</b> Authored comprehensive technical documentation and a laboratory handbook to ensure smooth knowledge transfer and onboarding for future developers.</li></ul> |                        |

## PROJECTS

### Deep Learning Algorithms Implementation (Python/Julia)

- **Algorithm Implementation:** Reproduced complex Deep Learning algorithms from C. Bishop's *Foundations and Concepts* to understand the low-level optimization mechanics of neural networks.
- **Data Visualization:** Developed interactive visualizations using Python (Matplotlib) and Julia to display model convergence and loss metrics, similar to dashboard reporting.
- **Version Control:** Maintained a clean Git history with structured commits and documentation, demonstrating proficiency in version control workflows.

## TECHNICAL SKILLS

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|--------------------------|------------------------------------------------------|
| <b>Core Languages:</b>   | Python (Pandas, NumPy, SciPy), SQL, Julia, C++, Bash |
| <b>Data Engineering:</b> | Data Pipelines, ETL Concepts, Git/GitHub, Linux/Unix |
| <b>Machine Learning:</b> | PyTorch, TensorFlow, OpenCV, Scikit-learn            |
| <b>Visualization:</b>    | Matplotlib, Data Visualization Principles            |
| <b>Concepts:</b>         | Object-Oriented Programming (OOP), REST APIs, Agile  |
| <b>Languages:</b>        | English (Fluent), Mongolian (Native)                 |